Lab 5 - Static Code Analysis

Name: Samyak Sanklecha
SRN: PES2UG24CS436

Repo: L5_02_retail_pos (pos_system.py)

GitHub: https://github.com/samyak-1021/SE_Lab_PES2UG24CS436/tree/main/Lab5

Before

Running it, bill 3 is -630 and bill 4 is 0

```
Bill 1: 126.0
Bill 2: 838.95
Bill 3: -630.0
Bill 4: 0
After promo: 301.45500000000004
Enter promo code (e.g. total - 50):
After your promo: 284.95000000000005
Sales so far: 334.95000000000005
(.venv) (base) samyaksanklecha@Samyaks-MacBook-Air L5_02_retail_pos %
```

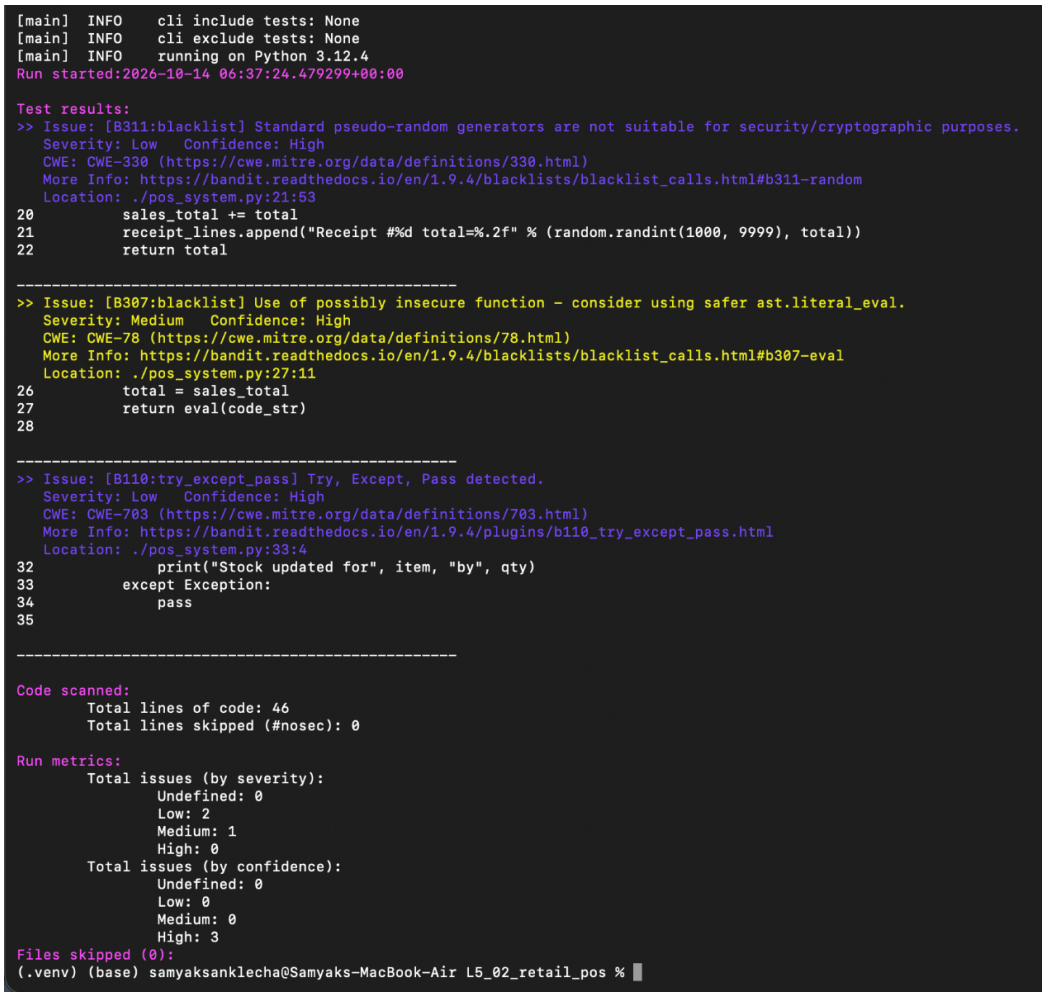
Pylint 5.65/10

```
************* Module pos_system
pos_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
pos_system.py:7:0: C0103: Constant name "sales_total" doesn't conform to UPPER_CASE naming style (invalid-name)
pos_system.py:10:0: C0116: Missing function or method docstring (missing-function-docstring)
pos_system.py:10:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
pos_system.py:11:4: W0603: Using the global statement (global-statement)
pos_system.py:16:8: W0702: No exception type(s) specified (bare-except)
pos_system.py:21:25: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
pos_system.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
pos_system.py:27:11: W0123: Use of eval (eval-used)
pos_system.py:26:4: W0612: Unused variable 'total' (unused-variable)
pos_system.py:29:0: C0116: Missing function or method docstring (missing-function-docstring)
pos_system.py:33:11: W0718: Catching too general exception Exception (broad-exception-caught)
pos_system.py:36:0: C0116: Missing function or method docstring (missing-function-docstring)
pos_system.py:37:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
pos_system.py:37:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
pos_system.py:41:0: C0116: Missing function or method docstring (missing-function-docstring)
pos_system.py:41:0: C0103: Function name "printSales" doesn't conform to snake_case naming style (invalid-name)
pos_system.py:44:0: C0116: Missing function or method docstring (missing-function-docstring)
pos_system.py:1:0: W0611: Unused import json (unused-import)
pos_system.py:3:0: W0611: Unused import logging (unused-import)

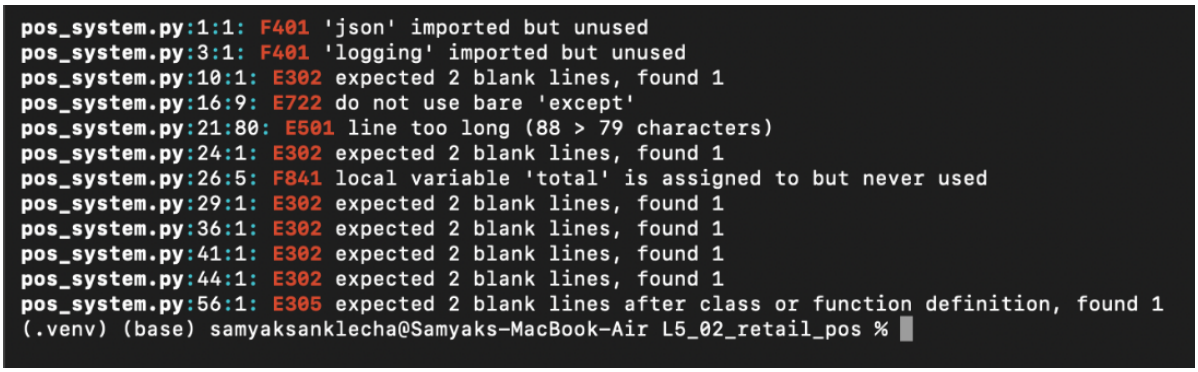
-----------------------------------
Your code has been rated at 5.65/10

(.venv) (base) samyaksanklecha@Samyaks-MacBook-Air L5_02_retail_pos %
```

Bandit, 3 issues



Flake8, 12 issues



Fixed: eval, bare except, negative qty/tax, mutable default arg, file handling, formatting.

After

Running it, bad bills get rejected now

```
Bill 1: 126.0
Bill 2 rejected: Unknown item: laptop
Bill 3 rejected: Invalid quantity for bottle: -4
Bill 4 rejected: Invalid tax rate: -1
Stock not updated: Unknown item: laptop
After promo: 113.4
Enter promo code (e.g. total - 50):
After your promo: 76.0
Sales so far: 126.00
(.venv) (base) samyaksanklecha@Samyaks-MacBook-Air Lab5 %
```

Pylint 10.00/10

```
-------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 5.65/10, +4.35)

(.venv) (base) samyaksanklecha@Samyaks-MacBook-Air Lab5 %
```

Bandit, no issues

```
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.12.4
Run started:2026-10-14 06:38:32.601922+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 92
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
(.venv) (base) samyaksanklecha@Samyaks-MacBook-Air Lab5 %
```

Flake8, nothing

```
flake8: no issues found
(.venv) (base) samyaksanklecha@Samyaks-MacBook-Air Lab5 %
```

Commits

```
3b76c03 (HEAD -> main, origin/main, origin/HEAD) Lab5: add after reports, issues table and reflection
c27f4e0 Lab5: fix logic bugs not caught by the tools (input validation, money rounding, stock update, duplicate log lines)
c8a4529 Lab5: fix Pylint issues (docstrings, global state, mutable default, with/encoding, naming) - 10.00/10
7f04f9d Lab5: fix Flake8 issues (blank lines, line length, unused imports, bare except)
aa065da Lab5: fix Bandit issues (B307 eval, B311 random, B110 try-except-pass)
ebe2285 Lab5: add original pos_system.py and baseline static analysis reports
6c6220d Add SE Lab 3 - component diagram and architecture justification
(.venv) (base) samyaksanklecha@Samyaks-MacBook-Air Lab5 %
```